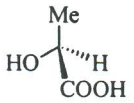
Organic Cumulative Examination

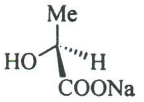
January 24, 2014

1) What is the difference between '*stereoisomers*' and '*constitutional isomers*'? Explain and give one example for each term. (15 points)

2) Is the following statement true or false: As compound I is named S-(+)-lactic acid, compound II has to be named S-(+)-sodium lactate. Explain your reasoning. (15 points)

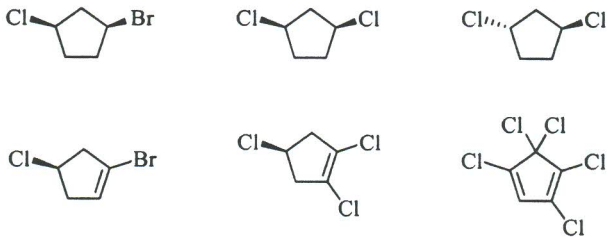


I

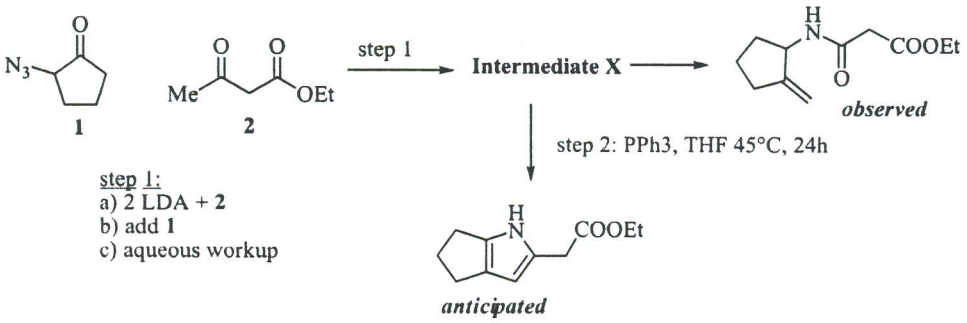


II

3) Which of the following compounds are chiral, which are not? Give a rational for your answer for each compound. (15 points)



4) The reaction given below documents an *errant* Staudinger reaction. The anticipated reaction product was pyrrole **3**, while the observed product was compound **4**. Identify the **<u>intermediate X</u>** and give a detailed reaction mechanism to the observed product (55 points)



<u>step 1</u>:
a) 2 LDA + **2**
b) add **1**
c) aqueous workup